

TripNest

Smart Travel Booking & Tour Management System

TripNest is a role-based full-stack travel booking web application developed using Spring Boot, React JS, Bootstrap, and MySQL. The system provides a digital platform for tourists and travel companies.

Technology Stack:

- Frontend: React JS, Bootstrap 5
- Backend: Spring Boot, Spring Security
- Database: MySQL
- ORM: Hibernate / JPA
- Testing: Postman

1. Project Overview

TripNest is designed to simplify travel package booking and tour management. Travel companies can add and manage packages while tourists can search, book, and manage trips online. The system uses role-based authentication for Admin, Company, and Tourist users.

2. Project Objectives

- Provide online travel booking platform
- Reduce manual travel management process
- Allow companies to manage travel packages digitally
- Allow tourists to search and book packages online
- Implement secure role-based authentication system

3. Main Modules

Module	Description
Admin Module	Manage users, companies, bookings, feedback, and packages.
Company Module	Add packages, trips, and manage bookings.
Tourist Module	Search packages, add wishlist/cart, and book trips.
Authentication Module	Login and registration with role-based access.
Booking Module	Handle trip booking and payment status.
Feedback Module	Store ratings and reviews from tourists.

4. Database Design According to ER Diagram

The database is designed using relational mapping between users, companies, packages, trips, bookings, wishlist, cart, and feedback. Each module is connected through primary and foreign keys.

Table Name	Purpose
role	Stores user roles such as Admin, Company, Tourist
users	Stores login and authentication information
company	Stores company details
package	Stores travel package details
trips	Stores trip schedule and travel information
tourist	Stores tourist profile details
booking	Stores booking information
addtocart	Stores cart details
addtowishlist	Stores wishlist items
feedback	Stores reviews and ratings

5. System Workflow

User Registration Flow:
User registers into the system and selects role (Tourist or Company).

Login Flow:

System verifies username and password and redirects user according to role.

Booking Flow:

Tourist searches package → Adds to cart/wishlist → Books trip → Payment status updated.

Company Flow:

Company adds packages and trips → Tourists view and book trips.

6. Advanced Features & Improvements

- JWT Authentication & Spring Security
- Role-Based Dashboard
- Search & Filter System
- Package Image Upload
- Booking Invoice PDF Generation
- Email Notifications
- Razorpay Payment Integration
- AI Travel Assistant Chatbot
- Booking Status Tracking
- Admin Analytics Dashboard
- Responsive Mobile-Friendly UI

7. REST API Design

Authentication APIs

POST /api/auth/register

POST /api/auth/login

Package APIs

GET /api/packages

POST /api/packages

PUT /api/packages/{id}

Booking APIs

POST /api/bookings

GET /api/bookings/user/{id}

Wishlist APIs

POST /api/wishlist

GET /api/wishlist/{id}

8. Future Scope

- Hotel Booking Integration
- Flight Ticket Integration
- Mobile Application Development
- AI-Based Trip Recommendation
- Multi-language Support
- GPS & Google Maps Integration

9. Conclusion

TripNest provides a complete digital travel booking and management solution. The project demonstrates frontend-backend integration, role-based authentication, CRUD operations, booking workflow, REST APIs, database relationship handling, and modern UI design. It is a strong and professional CDAC final project.